

# Comparisons of Scalar Implicatures and Free Choice Effects

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## Abstract

In this paper I compare scalar implicatures (SIs) and free choice (FC) effects in various respects, and claim that FC effects are not SIs. Intuitively, SIs involve the negation operator, while FC effects involve existential quantification. This indicates that their sources are different. This is empirically supported that in down-ward entailing (DE) contexts standard SIs do not arise. But FC effects are observed in some DE contexts. This means that FC effects do not come from monotonicity or scalarity. I attribute FC effects to the condition that every expression must make a meaning contribution to the meaning of the sentence it occurs independently of the other expressions. This leads to the existence of some semantic entities in a domain of a quantifier, normally the quantifier which has immediately wide scope over a disjunction structure. Therefore the same effects are expressed from uses of other expressions. Thus I also discuss similar effects from conjunction structures and conditionals.

## 1. Introduction

There are two phenomena which look so closely related to each other that people tend to see them as a kind: scalar implicatures (SIs) and free choice (FC) effects. One reason is that the same expression *or* gives rise to both of them. Among these two, the reasoning behind SIs is relatively well established. On the other hand, there are various attempts to explain how FC effects arise and there is not much agreement on the reasoning behind them. In this paper, I compare SIs and FC effects in various respects and capture the reasoning behind FC effects.

First, look at some examples of SIs:

- (1) John read some of the books.  
+> John did not read all of the books.
- (2) John took an exam or made a presentation.  
+> John did not (both) take an exam and make a presentation.

Basically SIs arise in the following general pattern:

### General Pattern of SIs:

$\phi \text{ +> } \neg\phi([A'/A])$ , where  
 $\phi([A'/A])$  is a sentence  $\phi$  with  $A$  in  $\phi$  replaced with  $A'$ , and  
 $(\dots, A', \dots, A, \dots)$  forms a scalar set.

By uttering  $\phi$ , the speaker implicates that a stronger statement than  $\phi$  is false, and the implicature is motivated by Grice's maxim of quantity and the existence of a relevant scalar set which includes an expression in  $\phi$ . Relevant scalar sets in (1) and (2) are (*all, most, many, some*) and (*and, or*), respectively.

Von Wright (1968) and Kamp (1973, 1979) observed that a disjunction sentence can be interpreted like a conjunction sentence.

- (3) You may eat an apple or a pear. ([MAY [A or B]])  
+> You may eat an apple and you may eat a pear. ([MAY A] and [MAY B])

In general *or* makes a weaker statement than *and*, but in this example a sentence with *or* seems stronger than one with *and*. This meaning relationship can be generalized in the following pattern:

General Pattern of FC interpretation:

$\phi(A \text{ OR } B) \rightarrow \phi(A) \text{ AND } \phi(B)$ , where

$\phi$  involves an *existential operator* which has wide scope over *or*.

In (3) *may* is an existential quantifier over deontic alternatives of the actual world, so the example fits with the general pattern. We can see examples in which other existential quantifiers are involved:

(4) **Some** students were sleeping or watching TV.

+> Some students were sleeping.

+> Some students were watching TV.

(5) John **might** be in the US or in Germany.

+> John might be in the US.

+> John might be in Germany.

In (4) *some* is an existential quantifier over individuals, and in (5) *might* is one over epistemic alternatives of the actual world.

The disjunction operator *or* forms a scalar set together with *and*, but it also gives rise to FC effects. To explain these two phenomena, some linguists propose an extended scalar set which includes a couple of null operators, which take a disjunction structure and give each disjunct in it. In their analyses, FC effects are just taken to be SIs. A direct motivation for claiming that FC effects are SIs is Alonso-Ovalle's (2005) observation that FC effects do not arise in downward entailing (= DE) contexts, just like any other SI (Kratzer & Shimoyama 2002, Alonso-Ovalle 2005, Fox 2006, Klinedinst 2006, etc.):

(6) a. His mother does not think that John may eat an apple or a pear.

b. His mother thinks that NOT [John may eat an apple or a pear] (Standard meaning)

= His mother thinks that John is not allowed to eat an apple and that he is not allowed to eat a pear (either).

c. His mother thinks that NOT [John may eat an apple and John may eat a pear] (Negation of the FC reading)

= His mother thinks that (either) John is not allowed to eat an apple or John is not allowed to eat a pear.

The standard meaning of sentence (6a) is (6b), in which no FC effect arises. If a FC effect arose, it would be interpreted as in (6c). Alonso-Ovalle attributes this to the fact that the disjunction structure lies in a DE context.

However, there is some evidence that free choice effects do arise in DE contexts:

(7) Every student who slept or watched TV failed the test.

+> Some students watched.

+> Some students watched TV.

On the other hand, a typical SI does not arise from the use of *or* in the same context. (7) does not convey the meaning as given in (8):

(8) ??Every student who slept or watched T, but did not do both, failed the test.

The question is whether FC effects are really SIs. In this paper, I will compare SIs and FC effects (which I call **existence implicatures**, or **EIs**) to show FC effects, or more generally EIs, are not SIs but that they are of the same kind as clausal implicatures. In Section 2, I will discuss differences between them. In Section 3, I will propose an analysis of EIs. Section 4 I discuss the common property of SIs and EIs: implicatures are always global. In Section 5, I point out that in accounting for EIs, we need quantifiers, and that a universal quantification is always involved in interpreting sentences. In Section 6, I discuss other than disjunction structures that also give rise to EIs. In doing so, I will show that EIs are quite general phenomena. In Section 7 I summarize the discussions in the paper.

## 2. Differences between EIs and SIs

### 2.1. Negation vs. existential quantification

The basic principle involved in calculating SIs is Grice's maxim of quantity:

Grice's (1957) Maxim of Quantity:

Make your contribution as informative as is required (for the purposes of exchange).

But he did not give the precise procedure of deriving a SI. Following his idea, Sauerland (2004) proposed a more precise procedure.

Sauerland's (2004) procedure:

If  $\psi \in \text{ScalAlt}(\phi)$  and  $\psi \Rightarrow \phi$  and not  $\phi \Rightarrow \psi$ , then  $\neg K\psi$  is a *primary implicature* of  $\phi$ . (Ignorance Inferences)<sup>1</sup>

If  $\neg K\psi$  is a primary implicature of  $\phi$  and  $K\neg\psi$  is consistent with the conjunction of  $\phi$  and all primary implicatures of  $\phi$ , then  $K\neg\psi$  becomes a *secondary implicature* of  $\phi$ .

The basic idea is that a scalar alternative of an asserted sentence leads to an implicature only if the scalar alternative is logically stronger (in the sense of asymmetric entailment) than the assertion. A more important thing to mention is the distinction between the "not-know" (= " $\neg K$ ") implicatures that directly follow from the Gricean maxim of quantity and the "know-not" (= " $K\neg$ ") implicatures Gazdar predicts, as given below. For the explanation of the implicatures from a disjunction structure, Sauerland introduces a terminological distinction between these two types of implicatures: he calls the " $\neg K$ " implicatures *primary implicatures*, and the " $K\neg$ " implicatures *secondary implicatures*. When a speaker *sp* utters a sentence  $\phi$ , and  $\psi$  is a stronger alternative of  $\phi$ , Grice's quantity maxim predicts that it is implicated that *sp* does **not know**  $\psi$  ( $\neg K\psi$ ). This weaker inference can be strengthened into a secondary implicature, which is that *sp* **knows** it is **not** the case that  $\psi$  (=  $K\neg\psi$ ), if the latter is compatible with  $\phi$  and all other primary implicatures.

We can apply this to a concrete example:

- (9) John read some of the books. (= (1))
- +> John did not read all of the books.
  - (all, most, many, some)
  - $\neg K(\text{John read all of the books})$
  - $K\neg(\text{John read all of the books})$

The ignorance inference is that  $\neg K(\text{John read all of the books})$ , and the strengthened proposition that  $K\neg(\text{John read all of the books})$  is compatible with the rest of the context. The secondary implicature is what is so-called a SI. And we can see that a SI is the negation of a stronger alternative statement, ignoring the epistemic operator *K*.

Typical FC effects arise when the disjunction operator occurs within the scope of an existential quantifier, but quantifiers other than an existential quantifier give rise to existence implicatures:

- (10) **Many/Most** students were sleeping or watching TV.
- +> **Some** students were sleeping.
  - +> **Some** students were watching TV.
- (11) **Everyone** ordered beer or soju.
- +> **Some** people ordered beer.
  - +> **Some** people ordered soju.
- (12) John **must** take an exam or give a presentation.
- +> John **may** take an exam.
  - +> John **may** give a presentation.
- (13) John **always/often** has coffee or juice at breakfast.
- +> John **sometimes** has coffee at breakfast.

<sup>1</sup> These conditions must be revised when the relationship between  $\phi$  and  $\psi$  is not strictly semantic. In discussing SIs, we should include cases where the scale is pragmatic. In this paper, however, I do not deal with such cases.

+> John **sometimes** has juice at breakfast.

In (10-11), the quantifiers are stronger scalar alternatives than existential quantifiers. In (12-13), the quantifiers quantify over possible worlds and times, respectively. If EIs were SIs, they would be derived by negating stronger alternative statements. EIs always involve **existential quantification**. This is why I call them **EIs**.

## 2.2. Maxim of quantity vs. maxim of relation

Behind the generation of SIs lies the maxim of quantity. In SIs, in general, a stronger alternative does not significantly change the complexity of the sentence. To account for EIs as SIs, Fox (2006) assume, following Sauerland (2004), that the relevant scalar set of *or* is (*and*, R, L, *or*), where R and L are operators which select the right and left elements in (A R/L B). But this assumption is not motivated in various respects, and R and L do not seem to be proper alternatives to *and* or *or*. First, they just select only one disjunct, so the resulting expressions differ in their complexity from the other two. That is, 'A' and 'B' are more simple than 'A or B' and 'A and B'. In addition, if a null element were considered a scalar alternative of *or*, almost every lexical item would constitute a scalar set with its corresponding null element.<sup>2</sup>

(14) a. Some small boxes are in the basement.

b. Some  $\emptyset$  boxes are in the basement.

(small,  $\emptyset$ )?, b +> a?

(big,  $\emptyset$ ), (blue,  $\emptyset$ ), ...

Here the adjective *small* is added to make the sentence stronger. Then uttering a sentence without it, like (14b), would implicate that NOT(some small boxes are in the basement). Given that a null element is an alternative to all adjectives like *small*, we can think of a lot of possible implicatures. We could also think of a null element as an alternative to all nouns like *boxes*. Moreover, there is no language which has an operator like R/L. For these reasons, the assumption of R and L are unmotivated.

If we assume that R and L are not scalar alternatives to *or*, then EIs are not SIs and they are generated from a different principle. Which maxim works in generating EIs? This is related to the question of why EIs always have existential quantification. I claim that what is relevant here is the **maxim of relation**. Adding an expression is relevant in a context when it makes a non-trivial meaning contribution in the context. EIs have the form of existential quantification. When we use a disjunction structure, each disjunct must make its own meaning contribution excluding the effect of the other disjunct. This predicts the following implicatures schematically:

(15) QUANT(R1 or R2)(N)

+> SOME(R1  $\wedge$   $\neg$ R2),

+> SOME(R2  $\wedge$   $\neg$ R1)

QUANT(R)(N1 or N2)

+> SOME(R  $\wedge$  N1  $\wedge$   $\neg$ N2),

+> SOME(R  $\wedge$  N2  $\wedge$   $\neg$ N1)

The meaning contribution of each disjunct is made by leaving some semantic entities in the domain of a quantifier. The use of *or* makes a sentence weaker, which leaves in the context some semantic entities which are solely traceable to the use of each disjunct. The set of entities is the evidence of the *existence* of semantic contribution. This is the source of an EI and this explains why EIs have existential force. EIs arise on the basis of the maxim of relation. Each disjunct is relevant in the context only if they make their own meaning contribution in the context.

## 2.3. Symmetric behaviors of SIs vs. asymmetric behaviors of EIs

<sup>2</sup> Consider the following example:

The quantifier *no* is symmetric in the sense that a sentence of the form 'no A B' has the same truth-conditions as 'no B A'. But a disjunction structure in the restrictor always yields EIs, while one in the nuclear scope does not.

(16) No students who sang or danced went to the final.

+> Some students sang.

+> Some students danced.

(17) No students sang or danced.

+/> Some students sang.

+/> Some students danced.

I suppose that this is related to the fact that a quantifier can be interpreted as presuppositional. (de Jong and Verkuyl. 1985, Lapin and Reinhart 1988) In (16), the quantifier subject can be taken to presuppose that there is a set of students who sang or danced, and the sentence is about the set of students. Sentence (16) can be followed by a sentence with a pronoun referring to the set of students, like *They had to wait for a next opportunity*. In (17), on the other hand, the quantifier *no* asserts that there was no student who sang or danced. This is in direct conflict with potential EIs.

This is contrasted with SIs. Regardless of whether the disjunction structure occurs in the restrictor or in the nuclear scope, no SIs arise in comparison with *and*. If EIs were SIs, they would have to behave in the same way. So we need to see what constraints are involved in SIs and EIs.

#### 2.4. Restrictions on SIs and EIs

Chierchia (2002) claimed that SIs are calculated locally, but whether they really occur in a local context is sensitive to the monotonicity of the local context with respect to the global context. That is, a SI arises only if a scalar term in a sentence is replaced with a stronger alternative and the whole sentence gets stronger. That is, a SI arises only when it is in an upward entailing (= UE) context. Thus we can predict that if a scalar term is used in a non-monotonic or DE context, it does not yield a SI. One such context is the antecedent clause of a conditional, which introduces a non-monotonic context:

(18) a. If some guests found the taste of the stew strange or got sick, then the stew must have been spoiled.

(Eckardt, 2007)

b. ??If some guests found the taste of the stew strange or got sick but did not do both, then the stew must have been spoiled. (SI)

In the antecedent clause of the conditional, the use of *or* does not yield a SI. Thus (18a) is not understood as conveying the meaning of (18b). On the other hand, we still get the following EIs in the same context.

(18') +> It is possible that some guests found the taste of the stew strange. (EI)

+> It is possible that some guests got sick. (EI)

In some obviously DE contexts, no SIs arise, while EIs do.

(7) **Every** student who slept or watched TV failed the test.

≠+> Every student who slept or watched TV but did not do both failed the test. (SI)

+> Some students slept (but did not watch TV). (EI)

+> Some students watched TV (but did not sleep). (EI)

(19) **No** one who sang or danced went to the final.

≠+> No one who sang or danced but did not do both went to the final. (SI)

+> Some people are in the kitchen. (EI)

+> Some people are in the living room. (EI)

(20) {**Less than a hundred, Few**} students speak French or German.

≠?> {Less than a hundred, Few} students speak French or German but not both. (SI)

+> Some students speak French. (EI)

+> Some students speak German. (EI)

The quantifiers *no*, *less than a hundred*, and *few* introduce a DE restrictor and a DE nuclear scope. And the quantifier *every* introduces a DE restrictor. In these contexts, no SIs arise but EIs do.

On the other hand, there are cases where EIs are sensitive to monotonicity. We have seen that each disjunct in a disjunction structure makes a meaning contribution by leaving some semantic entities in the domain of a quantifier which has immediately wide scope over the disjunction operator. But when the negation operator intervenes between the disjunction operator and a quantifier, adding each disjunct does not leave any semantic entities in the domain of the quantifier, but eliminates some semantic entities related to each disjunct instead:

(21) Some women are not sleeping or watching TV.

For this reason in (21) no EIs arise.<sup>3</sup> This shows that EIs ARE sensitive to monotonicity, but only when there is an operator intervening between a disjunction structure and a quantifier making the context within the scope of the quantifier non-monotonic. This indicates that EIs arise with respect to a quantifier which has immediately wide scope over a disjunction structure.

Now a question arises why no EIs arise in (6). There is another restriction on EIs: plurality. As pointed out in Klinedinst (2006), when an expression gives rise to an EI, the entities in the domain of a quantifier which satisfy the expression must be **plural**:

- (22) a. In the cinema, some woman was sleeping or keeping her eyes closed.  
b. In the cinema, at least one woman was sleeping or keeping her eyes closed.

In these two examples, the singular NPs *some woman* and *at least one woman* allow the possibility that the set of women who make the sentence true can be single-membered. This prohibits an EI from arising in these examples, because the EIs from a disjunction structure require at least one entity for each disjunct.

This also explains why the following example does not yield an EI.

- (23) No students speak French or German.  
+> Some students speak French.  
+> Some students speak German.

The set of students who make the sentence true is the empty set, which does not satisfy the condition of plurality. This also explains why no EIs arise in (6):

- (24) His mother does not think that John may eat an apple or a pear. (= (6))  
=> His mother thinks that it is not the case that John may eat an apple or a pear.  
+> His mother thinks that John may not eat an apple or that John may not eat a pear. (EI)

The truth conditions of sentence (24) is as follows:

Truth-conditions of (24):

<sup>3</sup> Then it is expected that an expression which normally eliminates semantic entities will leave some semantic entities instead. This is exactly what we observe in (i):

- (i) Some women are not sleeping and watching TV.  
+> Some women are sleeping but not watching TV.  
+> Some women are watching TV but not sleeping.

In (i), if the conjunct *sleeping* makes any semantic contribution, it leaves some women who are sleeping but not watching TV in the domain of the quantifier *some women*. Similarly, the conjunct *watching TV* leaves some women who are watching TV but not sleeping. This shows that the negation operator is different from other operators which introduce DE contexts. Quantifiers can yield EIs from the use of a disjunction structure, whether or not they introduce DE contexts. But the negation operator prohibits a disjunction structure from yielding EIs. I suppose it is because EIs are based on a domain of quantification and the negation operator is not a quantifier. A quantifier introduces a domain for yielding an EI, but the negation operator just changes the way an expression makes a meaning contribution to a domain of quantification.

For every epistemic alternative  $w$  of John's mother, NOT(there are some deontic alternatives of  $w$  in which John eats an apple or a pear).

= For every epistemic alternative  $w$  of John's mother, there are no deontic alternatives of  $w$  in which John eats an apple or a pear.

Here (24) asserts that the set of deontic alternatives of  $w$  in which John eats an apple or a pear is the empty set, which does not satisfy the condition of plurality. Potential EIs require at least one deontic alternative for each disjunct.

## 2.5. Overt expressions or implicit (stronger) alternatives

A SI arises due to the existence of a stronger alternative of the actually used expression, but the stronger alternative is given by the language system, not by the speaker herself. An EI, on the other hand, arises due to an actually used expression.

SI: the negation of (part of) a sentence with an expression replaced with an implicit stronger alternative

EI: an existential quantification over some domain introduced by some overt expression in the sentence

(25) John read some of the books.

ALT: John read all of the books.

+> John did **not** read all of the books.

(26) **Many** students were sleeping or watching TV.

+> **Some** students were sleeping.

+> **Some** students were watching TV.

In (25), the scalar alternative *all* is provided by the system of language. Assuming that *sp* is responsible for only the expressions in a sentence, *sp* is not overtly committed to its alternative statement or the negation of the alternative statement. A SI is just what *hr* supposes. In (26), on the other hand, an EI arises when *hr* assumes that *sp* is committed to the existence of some evidence for using each of the two disjuncts, and the two disjuncts are uttered by *sp*. *sp* is overtly committed to an EI. This is related to the next difference.

## 2.6. Cancelability

SIs are easily cancelled but EIs are not. In (27), the speaker cancels the implicature herself without making the sentence odd. This can be contrasted with (28), in which *sp* cancels the EI and the sentence becomes odd.

(27) Some people ordered juice or coffee or both.

SI: NOT(some people ordered both) = No one ordered juice and coffee.

(28) ??Some people ordered juice or coffee, but no one ordered juice.

EIs: (i) Some people ordered juice, (ii) some people ordered coffee.

SIs are based on nothing but ignorance inferences, which *sp* is not committed to, and it is also indeterminate whether ignorance inferences can become SIs or not. SIs are just hypothetical inferences on the part of a hearer, so *sp* can claim that they are false. In this sense, *sp* does not really cancel SIs, but makes a clarification on the indeterminate issue. On the other hand, EIs are *sp*'s overt commitments to the meaning contributions of the expressions she decides to use. Denying them is like a self-contradiction. This is why *sp* cannot deny EIs easily. This is also confirmed by an experimental study by Chelma (2009).<sup>4</sup>

<sup>4</sup> Sauerland (2009) cites Chelma's (2009) experiment in which subjects are given the following two types of premises, and are asked whether they get the inferences given:

- |      |                                                                                 |      |
|------|---------------------------------------------------------------------------------|------|
| (i)  | a. Premise: Everyone passed most of his exams.                                  |      |
| b.   | Inference tested: No one passed them all.                                       | (SI) |
| (ii) | a. Premise: Everybody is allowed to give me the dissertation or the commentary. |      |

### 3. EIs as meaning contributions of expressions

#### 3.1. Meaning contributions through semantic entities

The maxim of relation requires that every expression in a sentence make a non-trivial meaning contribution to the meaning of the whole sentence in a context, unless they are used solely for structural reasons.<sup>5</sup> Meaning contributions can be captured by semantic entities. I will explain how they can be.

I assume that a context is represented as a set of possible worlds which are compatible with the current information state. And each possible world has a set of individuals as its dwellers and various quantifiers quantify over possible worlds or individuals in relevant domains.

A context  $\sigma = \{w \in W \mid w \text{ is compatible with the information in an information state}\}$

For each possible world  $w \in W$ , there is a set of individuals  $D_w$ .

Uttering a sentence in a context changes it into a new one as follows:

Update of a context  $\sigma$  by a sentence  $\phi$ :

$\sigma[\phi]\sigma'$  ( $\sigma'$ : the input context,  $\sigma' = \{w \in \sigma \mid \phi \text{ is true in } w\}$ .)

Here  $\sigma'$  is a smaller set of possible worlds than  $\sigma$ , because possible worlds which are not compatible with  $\phi$  are excluded from  $\sigma$ .

If an expression is used in a sentence, its meaning contribution is captured by a set of semantic entities like possible worlds or individuals in a domain of quantification:

$$E = D \cup W$$

$D$  is a set of individuals in a possible world and  $W$  is a set of possible worlds. Semantic entities are individuals or possible worlds. I need to show how a meaning contribution is captured by a set of entities which is part of the domain.

To calculate the meaning contribution of an expression, we have to compare the sentence itself with the same sentence with that expression replaced with the semantically null element which corresponds to the expression. To do this, I have to first define null elements. What a null element of an expression is depends on the embedding structure in which it is used. I confine my discussion to conjunctive and disjunctive structures.

Null elements in conjunctive structures:

$$\begin{aligned} 1 & \langle e, t \rangle = \lambda d \in D . 1 \\ 1 & \langle e, \langle e, t \rangle \rangle = \lambda d \in D . [\lambda d \in D . 1] \\ 1 & \langle \langle e, t \rangle, \langle \langle e, t \rangle, t \rangle \rangle = \lambda f \in D_{\langle e, t \rangle} . [\lambda g \in D_{\langle e, t \rangle} . 1] \\ \dots & \end{aligned}$$

Null elements of disjunctive structures:

$$\begin{aligned} 0 & \langle e, t \rangle = \lambda d \in D . 0 \\ 0 & \langle e, \langle e, t \rangle \rangle = \lambda d \in D . [\lambda d \in D . 0] \\ 0 & \langle \langle e, t \rangle, \langle \langle e, t \rangle, t \rangle \rangle = \lambda f \in D_{\langle e, t \rangle} . [\lambda g \in D_{\langle e, t \rangle} . 0] \\ \dots & \end{aligned}$$

I will call these two a 1-element and 0-element respectively, and a Null element together. And the semantic contribution of an expression  $\alpha$  in a larger expression  $\phi$  is the difference between  $\phi$  and  $\phi[\alpha/\text{Null}_{\tau(\alpha)}]$ ,  $\phi[\alpha/\text{Null}_{\tau(\alpha)}]$  is  $\phi$  with the meaning of  $\alpha$  replaced with its corresponding Null element. If an expression is

b. Inference tested: Everybody can choose which of the two he will give to the teacher. (EI)

In (i) (*all, most, many, some*) is a relevant scalar set, and the inference is a SI. In (ii), the disjunction structure allows both options to be available as EIs. From the experiment, Chelma concludes that SIs are judged to be less strong than EIs.

<sup>5</sup> We can see some examples of semantically null for structural reasons:

(i) John is proud of his father.

Here we can assume that *is* and *of* have no semantic meanings except that *is* conveys the information of a time with respect to the utterance time. They are used in the sentence just because of grammatical reasons: *be* is necessary to carry a tense and *of* is motivated by the case-marking of the NP *his father*. The other expressions than these must convey some non-null meanings.



replaced with its corresponding null element, the meaning of the sentence is determined only based on the rest of the sentence.

- (29) a.  $\llbracket [a \text{ smart and pretty student}] [\text{smart}/1_{\langle e,t \rangle}] \rrbracket = \llbracket a \text{ pretty student} \rrbracket$   
 b.  $\llbracket [a \text{ smart or pretty student}] [\text{smart}/0_{\langle e,t \rangle}] \rrbracket = \llbracket a \text{ pretty student} \rrbracket$

I will represent a null element as  $\text{Null}_{\tau}$ , where  $\tau$  is a semantic type, and in actual calculation  $1_{\langle e,t \rangle}$  or  $0_{\langle e,t \rangle}$  will be used. Which one is used in an actual calculation is determined by the linguistic context.

Now we can define the meaning contribution by an expression in a sentence. When an expression  $\alpha$  occurs in the restrictor of a quantifier, it makes a meaning contribution by changing the domain of the quantifier. On the other hand, when an expression occurs in the nuclear scope of a quantifier, it makes a meaning contribution by separating things that are attributed only to that expression from the rest in the domain of the quantifier. This can be formally expressed as follows:

Given a quantificational structure  $\text{Op}(R)(N)$ , where  $\text{Op}$  is a quantifier,  $R$  the restrictor and  $N$  the nuclear scope,

Meaning contribution of an expression  $\alpha$ :

- (i)  $\alpha$  is in  $R$ :  $\{e \in E \mid R(e)\} \uparrow \{e \in E \mid R[\alpha/\text{Null}_{\tau(\alpha)}](e)\}$   
 (ii)  $\alpha$  is in  $N$ :  $\{e \in E \mid R(e) \wedge N(e)\} \uparrow \{e \in E \mid R(e) \wedge N[\alpha/\text{Null}_{\tau(\alpha)}](e)\}$   
 $A \uparrow B = [A \cup B] - [A \cap B] \quad (A \subseteq B \text{ or } B \subseteq A)$

( $P$  is the interpretation of an expression  $P$ ;  $\tau(P)$  is the semantic type of an expression  $P$ )

Here  $A \uparrow B$  represents the difference between the two sets  $A$  and  $B$ , when one set is a subset of the other. When the restrictor  $R$  contains an expression  $\alpha$ , the meaning contribution of  $\alpha$  is the difference of the set of entities which have the property of  $R$  and that of entities which have the property of  $R[\alpha/\text{Null}_{\tau(\alpha)}]$ , which is the restrictor we get by replacing  $\alpha$  with a null element. The null element is a 1-element or 0-element of the same semantic type as  $\alpha$ , depending on the linguistic context. Similarly, the meaning contribution of  $\alpha$  in the nuclear scope is captured only by the entities in the domain of the quantifier: the difference between the set of entities which have the properties of  $R$  and  $N$  and that of entities which have the properties  $R$  and  $N[\alpha/\text{Null}_{\tau(\alpha)}]$ . An expression in a nuclear scope reflects its meaning contribution through the domain of the quantifier because of the property of conservativity, that is, that the nuclear scope of a quantifier is interpreted with respect to a set of entities the restrictor provides.

### 3.2. Immediately wide scope over the disjunction operator

We are primarily concerned with the meaning contributions of a disjunction structure. In a sentence with a disjunction structure, a disjunct normally leaves a set of entities in the domain of a quantifier which has wide scope over the disjunction structure. This is the EI from the use of that disjunct. A disjunction structure can occur in the restrictor or in the nuclear scope, and the way a disjunct makes a meaning contribution is dependent on the location of the disjunction structure. First consider the case where a disjunction structure occurs in the restrictor of a quantifier.

For a quantificational structure  $\text{Op}(R1 \text{ or } R2)(N)$ ,

Meaning contribution of  $R1$ :

$$\{e \in E \mid R1(e) \wedge R2(e)\} \uparrow \{e \in E \mid \text{Null}_{\tau(R1)}(e) \wedge R2(e)\}$$

$$= \{e \in E \mid R1(e) \wedge \neg R2(e)\} \neq \emptyset$$

→ There are some  $e$  such that  $R1$  (but not  $R2$ ).

Meaning contribution of  $R2$ :

$$\{e \in E \mid R2(e) \wedge \neg R1(e)\} \neq \emptyset$$

→ There are some  $e$  such that  $R2$  (but not  $R1$ ).

The meaning contribution of a disjunct  $R1$  is the difference between the set of entities which have the properties of  $R1$  or  $R2$  and that of entities which have the property of  $R2$  since  $R1$  is replaced with its corresponding null element. It boils down to the set of entities which have the properties of  $R1$  but  $\neg R2$ . Here “ $\neg R2$ ” is added because the meaning contribution must be exclusively due to  $R1$ . If  $R1$  is to make a

non-trivial meaning contribution, the difference may not be the empty set. The non-trivial meaning contribution can be paraphrased as saying that there are some entities such that R1 but not R2. This is the EI from the use of R1. Likewise, the meaning contribution of the disjunct R2 is the set of entities which have the property of R2 but not R1, and the EI from R2 is that there are some entities such that R2 but not R1.

Consider the case where a disjunction structure occurs in the nuclear scope.

For a quantificational structure  $Op(R)(N1 \text{ or } N2)$ ,

Meaning contribution of N1:

$$\{e \in E \mid R(e) \wedge [N1(e) \vee N2(e)]\} \uparrow \{e \in E \mid R(e) \wedge [\text{Null}_{\tau(N1)}(e) \vee N2(e)]\}$$

$$= \{e \in E \mid R(e) \wedge N1(e) \wedge \neg N2(e)\} \neq \emptyset$$

→ There are some R such that N1 (but not N2).

Meaning contribution of N2:

$$\{e \in E \mid R(e) \wedge N2(e) \wedge \neg N1(e)\} \neq \emptyset$$

→ There are some R such that N2 (but not N1).

The meaning contribution of N1 is the difference between the set of entities which have the properties of the restrictor and the nuclear scope and that of entities which have the properties of the restrictor and N2 since N1 is replaced with its corresponding null element. It is the set of entities, among the entities in the domain of the quantifier, which have the properties of N1 and  $\neg N2$ . Thus we get the EI that there are some R such that N1 but not N2. We can get the meaning contribution, thus the EI, of N2 in the same way.

Now let's consider a concrete example in which a disjunction structure occurs in the restrictor of a quantifier.

(7) Every student who slept or watched TV failed the test.

+> Some students slept (but did not watch TV).

+> Some students watched TV (but did not sleep).

The quantifier *every student who slept or watched TV* introduces a domain of quantification and each disjunct makes a meaning contribution to it as follows:

$$\text{Domain of quantification: } \{d \in D \mid \underset{w}{student}(d) \wedge [\underset{w}{slept}(d) \vee \underset{w}{watched\_TV}(d)]\}$$

The meaning contribution of *passed an exam*:

$$\{d \in D \mid \underset{w}{student}(d) \wedge \underset{w}{slept}(d) \wedge \neg \underset{w}{watched\_TV}(d)\} \neq \emptyset$$

→ Some people passed an exam (but did not make a presentation).

The meaning contribution of *made a presentation*:

$$\{d \in D \mid \underset{w}{student}(d) \wedge \neg \underset{w}{slept}(d) \wedge \underset{w}{watched\_TV}(d)\} \neq \emptyset$$

→ Some people made a presentation (but did not pass an exam).

The disjunct *slept* makes a non-trivial meaning contribution by the non-empty set of students who slept but did not watch TV. It means that some students slept but did not watch TV. Similarly, the disjunct *watched TV* gives rise to the EI that some students watched TV but did not sleep.

Next, look at how a disjunction structure makes a semantic contribution in the nuclear scope of a quantifier.

(30) Everyone ordered beer or soju.

+> Some people ordered beer.

+> Some people ordered soju.

The quantifier *everyone* introduces a domain of quantification, and each disjunct makes a non-trivial meaning contribution and gives rise to an EI, as follows:

$$\text{Domain of quantification: } \{d \in D \mid \underset{w}{person}(d)\}$$

The meaning contribution of a disjunct *ordered beer*:

$$\{d \in D \mid \underset{w}{person}(d) \wedge \underset{w}{ordered\_beer}(d) \wedge \neg \underset{w}{ordered\_soju}(d)\} \neq \emptyset$$

→ Some people ordered beer (but did not order soju).

The meaning contribution of a disjunct *ordered soju*:

$$\{d \in D_w \mid \text{person}_w(d) \wedge \neg \text{ordered\_beer}_w(d) \wedge \text{ordered\_soju}_w(d)\} \neq \emptyset$$

→ Some people ordered soju (but did not order beer).

Thus the disjunct *ordered beer* yields the EI that some people ordered beer (but did not order soju), and the disjunct *ordered soju* the EI that some people ordered soju (but did not order beer).

#### 4. EIs and SIs are both local and global

Chierchia (2002) proposed that SIs be calculated locally in the process of semantic interpretation. He was not quite correct in his proposal, but it is clear that SIs can arise locally.

- (31) Some students who watched TV *or* played games failed maths.
- a. NOT[Some students who watched TV and played games failed maths]
  - b. Some students who NOT[watched TV and played games] failed maths.

In (31), the use of *or* gives rise to a SI globally, as in (31a), or locally, as in (31b), but the more plausible one is the latter, in which the negation operator applies in the relative clause.

There are also cases where global negation leads to a more appropriate SI.

- (32) Some students who drank some of the beer were allowed to drive.
- a. Some students who NOT[drank all of the beer] were allowed to drive.
  - b. NOT[some students who drank all of the beer were allowed to drive]
- (= No students who drank all of the beer were allowed to drive.)
- (33) Every student read some of the stories.
- a. Every student did NOT read all of the stories.
- (= No students read all of the stories.)
- b. NOT every student read all of the stories.

Here (32b) and (33b) are each preferred to (32a) and (33a). This shows that SIs can arise with the negation operator having various scopes.

However, even when SIs are derived locally, the calculation of a SI still has a characteristic of global calculation. The local context must be in an UE context, and this is always determined globally. In (31) the SI is calculated locally, but if the disjunction structure is embedded in the scope of the quantifier *no*, as in (34), then no SI arises.

- (34) No students who watched TV or played games failed maths.
- +/> NOT[NO students who watched TV and played games failed maths]
  - +/> No students who NOT[watched TV and played games] failed maths.

This shows that calculation of SIs is basically global, even if you can get a contextually appropriate SI locally.

EIs can also be calculated locally.

- (35) John must/may have an apple or a pear.
- +> John may have an apple.
  - +> John may have a pear.

We interpret the disjunction operator as having narrower scope than the deontic operator. Thus we get two EIs of permission. Existential quantification applied to the domain of the deontic quantifiers.

However, we can get EIs from global calculation. In (36), existential quantification applies globally, which leads to two EIs of epistemic possibility.

- (36) John must/may have an apple or a pear, but I don't know which.
- +> It is possible that John must/may have an apple.
  - +> It is possible that John must/may have a pear.

This shows that EIs also arise with the existential quantifier having various scopes. This can be attributed to the scope relationship in which the disjunction operator is taken to have wide scope over the deontic operators. In (36), the disjunction structure has wide scope over the deontic operators.

We have seen that SIs can be taken to be global in that they are derived only in UE contexts. What restriction is relevant in deriving EIs? I have claimed that EIs are meaning contributions to the domain of a quantifier which has wide scope over a disjunction structure. What if such meaning contributions are not guaranteed to remain in the main context? It would make each disjunct not contributive to the main context. Compare the two sentences below:

- (37) a. It is possible that everyone sang or danced.  
       +> It is possible that some people sang.  
       +> It is possible that some people danced.  
       b. It is impossible that everyone sang or danced.  
       +> It is possible that some people sang.  
       +> It is possible that some people danced.

In (37a), each disjunct makes a meaning contribution locally and the EI we get is that it is possible that some people sang and that some people danced. In (37b), on the other hand, we cannot derive EIs in the same way. If the existential quantifier applies to the local context, we might get the EI that it is impossible that some people sang and that some people danced. Here the operator *it is impossible that ...* nullifies the meaning contribution of each disjunct because it would eliminate from the context possible worlds in which some people sang and those in which some people danced. If each disjunct makes any meaning contribution, there must be some possible worlds remaining in the context in which some people sang and some in which some people danced. Therefore if we are to get an EI, it should be that it is possible that some people sang and that some danced. This shows that even if an EI arises in a local context, its effects have to go up to the main context, if it is compatible with the assertion. That is a global restriction on EIs. This condition can be satisfied even by the existence of possibility of such an effect, as shown in (37b).<sup>6</sup>

We can apply the same reasoning to (38a). If local EIs are combined with the meaning of the rest, we would get the meaning that John does not believe that some/any student sang/danced. But (38b,c) are more likely to be the inference we get from (38a).

- (38) a. John does not believe that most students who sang or danced went to the final.  
       b. John believes that (it is possible that) there were some students who sang.  
       c. John believes that (it is possible that) there were some students who danced.

This implies that local EIs are projected unless they are incompatible with the meanings of sentences.

## 5. Support conditions involved in EIs

Zimmermann (2000) and Geurts (2005) assumed that *or* is inherently an epistemic modal operator. One crucial reason is that epistemic inferences are almost always available even when there is no overt epistemic operator involved:

- (39) a. John is out or asleep.  
       +> John might be out (but not asleep).

<sup>6</sup> This allows us to account for a slightly different understanding of (7). The primary understanding of this sentence is that it implicates that there are some students who slept and some who watched TV. But still there is also the possibility that the speaker does not know for sure if there are some students who slept or if there are some who watched TV. Even in such cases, there must be some possibility of the existence of some students who slept and some who watched TV. This leads to the following EIs:

- (i) It is possible that there are some students who slept.  
       It is possible that there are some students who watched TV.

This reminds us of the comparison of ignorance inferences and SIs. The former are simply possibilities ( $\neg K\phi = P\neg\phi$ ). When these are strengthened into  $K\neg\phi$ , they become SIs. We can see the same variations in strengths.

- +> John might be asleep (but not out).  
b. John must/may have an apple or a pear, but I don't know which.  
+> It is possible that John must/may have an apple.  
+> It is possible that John must/may have a pear.

In (39a), there is no overt modal possibility operator, but from the two disjuncts we can get the corresponding EIs. In (39b), *must/may* is used as a deontic modal operator and due to the expression *I don't know which*, each disjunct is understood as an epistemic possibility.

This does not mean that the disjunction operator has the meaning of epistemic possibility inherently. In the following two examples, the disjunction operator can be understood as involving no epistemic possibility.

- (40) a. Everybody ordered a beer or a pizza. (Eckardt 2007)  
b. John is in love or I am a monkey's uncle. (Simons 2005)

Sentence (40a) can be assumed to be uttered in a situation the speaker knows who ordered what, as Eckardt (2007) pointed out. In such a situation there is no epistemic possibility involved. In (40b), the disjunction operator is simply truth-functional. The speaker is apparently not a monkey's uncle, and it is simply asserted that John is in love. These examples show that epistemic possibility in other uses of disjunction structures is simply an implicature, not inherent part of the semantics of the disjunction operator.

In my analysis, an epistemic meaning of the disjunction operator should be analyzed as a meaning contribution to a domain of quantification over epistemic alternatives. This is what I am going to do. Suppose the sentence is uttered in a context  $\sigma$ . Then the meaning of a sentence is a context change potential, and the context  $\sigma$  changes into a new context  $\sigma'$ . The new context supports the sentence since the following condition is satisfied:

$$\sigma\{\phi\}\sigma'$$

$\phi$  is supported in the updated context  $\sigma'$  iff for **every** possible world  $w$  in  $\sigma'$ ,  $\phi$  is true in  $w$ .

If a context is updated with a sentence, the sentence is true in every possible world in the resulting context. That is, the condition on support is universal quantification over possible worlds in the resulting context. Thus (39a) is supported in a resulting context  $\sigma'$ , as follows:

For every possible world  $w$  in  $\sigma'$ , John is out in  $w$  or John is asleep in  $w$ .

The support condition introduces a universal quantifier over possible worlds, and each disjunct in a sentence can make a meaning contribution to the domain of the quantifier. These should not be empty if each disjunct is used felicitously, as given below:

- Meaning contribution of *out*:  
 $\{w \in \sigma' \mid \text{John is out in } w, \text{ John is not asleep in } w\} \neq \emptyset (= \text{John might be out.})$   
Meaning contribution of *asleep*:  
 $\{w \in \sigma' \mid \text{John is asleep in } w, \text{ John is not out in } w\} \neq \emptyset (= \text{John might be asleep.})$

These conditions are just EIs.

We can get similar EIs in (39b) too. A obligation/permission statement with the disjunction operator can have two interpretations depending on the scope of the disjunction operator:<sup>7</sup>

- (39b') John must/may have an apple or a pear  
The sentence is supported in  $\sigma'$  iff  
(i) for (i)**every** possible world  $w$  in  $\sigma'$ , for (ii)**every/some**  $w'$  in  $\text{ACC}_D(w)$ , John have an apple **OR** a pear in  $w'$ ; (must/may > or)

<sup>7</sup> The scope of the disjunction operator is assumed to be determined syntactically, following Larson (1985) and Schwarz (1999).

- (ii) for (i)every possible world  $w$  in  $\alpha'$ , for (ii)every/some  $w'$  in  $ACC_D(w)$ , John have an apple **OR** for (ii)every/some  $w''$  in  $ACC_D(w)$ , John have a pear in  $w''$ . (or  $>must/may$ )  
( $ACC_D(w)$  is a set of deontic alternatives of a possible world  $w$ .)

In (39b'i), the disjunction operator has narrower scope than the deontic operators, so each disjunct makes a meaning contribution to the domain of deontic alternatives. Thus we get the following two EIs:

- (41) John may have an apple.  
John may have a pear.

In (39b'ii), on the other hand, the disjunction operator has wide scope over the deontic operators. In this case, each disjunct makes a semantic contribution to the universal quantifier over epistemic alternatives in  $\alpha'$ . Thus we get the following EIs:

- (42) It is possible that John must/may have an apple.  
It is possible that John must/may have a pear.

Sentence (39b) yields these EIs because of the expression *I don't know which*. It requires the scope relation of (40b'ii).<sup>8</sup>

Fox (2006) and Klinedinst (2006) assume a syntactically covert operators like *Exh*(austive operator) and *STRONG* (which includes the meaning of distributive operator and scalar implicature) respectively. Their syntactic approach do not have access to the universal quantification over the epistemic alternatives in an information state. Thus a syntactic approach must resort to a separate process in accounting for cases where a disjunction structure has the widest scope.

## 6. Other cases of EIs

### 6.1. Conjunction structures

EIs arise when an additional expression leaves a set of entities which are attributed only to that expression. In a disjunction structure a disjunct is such an expression. In a conjunction structure, on the other hand, adding a conjunct makes the statement stronger and excludes a set of entities which are attributed only to that expression. If such an expression occurs in the restrictor of a quantifier, it normally narrows down the domain of a quantifier by eliminating some entities from the domain. Hence no EIs. An example is given here:

- (43) Every student who slept and watched TV failed the test.

Adding *slept* reduced the quantification domain from a set of students who watched TV to a set of students who both watched TV and slept. We can get a similar effect from adding *made a presentation*.

If the conjunction structure is in the scope of the negation operator, however, a conjunct can generate an EI by leaving some entities in the domain which have one of the two properties due to each conjunct.<sup>9</sup>

<sup>8</sup> Simons (2005) pointed out when a sentence includes an epistemic operator, it is odd to add "... but I don't know which". On the other hand, when an epistemic statement is based on what the speaker is not responsible for, it is fine to add the expression:

- (i) ??John must/might be our or asleep, but I don't know which.  
(ii) Someone said: The fugitive is in Cambridge.  
I say: The fugitive might be in Massachusetts or in England, but I don't know which.

This can be explained by the property that an epistemic operator is a test of the information state in dynamic semantics, as Groenendijk, Stokhof and Veltman (1996) claim and the epistemic operators must be interpreted with respect to an agent whose information state is involved, as Yeom (1997) claim.

<sup>9</sup> Fox (2006) also gives an example of this:

- (44) a. Not many students passed an exam (= PE) and made a presentation (MP).  
 b. Many students did not pass an exam and make a presentation.
- (45) +> Some people did not pass an exam but made a presentation.  
 +> Some people did not make a presentation but passed an exam.

Whether the negation operator has scope over a quantifier or not, the conjunction structures in (44a) and (44b) yield EIs. I claim that this is related to the fact that the two quantifiers *not many* and *many* introduce the same set of entities and some of them have a property related to one of the two conjuncts, regardless of the scope of the negation operator. In both examples, when we calculate the meaning contributions of the two conjuncts, we find that they are the same:

(46) Meaning contribution of a conjunct *pass an exam* (=PE) in (56a):

For every possible world  $w \in \mathcal{O}'$ ,  $\{d \in D_w \mid person_w(d) \wedge PE_w(d) \wedge MP_w(d)\}^\dagger$   
 $\{d \in D_w \mid person_w(d) \wedge Null_{\tau(PE)}^w \wedge MP_w(d)\}$   
 $\{d \in D_w \mid person_w(d) \wedge \neg PE_w(d) \wedge MP_w(d)\} \neq \emptyset$   
 → Some people did not pass an exam but made a presentation.

Meaning contribution of a conjunct *made a presentation* (=MP) (47a):

For every possible world  $w \in \mathcal{O}'$ ,  $\{d \in D_w \mid person_w(d) \wedge \neg MP_w(d) \wedge PE_w(d)\} \neq \emptyset$   
 → Some people did not make a presentation but passed an exam.

(47) Meaning contribution of a conjunct *pass an exam* in (47b):

For every possible world  $w \in \mathcal{O}'$ ,  $\{d \in D_w \mid person_w(d) \wedge \neg[PE_w(d) \wedge MP_w(d)]\}^\dagger$   
 $\{d \in D_w \mid person_w(d) \wedge \neg[Null_{\tau(PE)}^w \wedge MP_w(d)]\}$   
 $\{d \in D_w \mid person_w(d) \wedge \neg PE_w(d) \wedge MP_w(d)\} \neq \emptyset$   
 → Some people did not pass an exam but made a presentation.

Meaning contribution of a conjunct *made a presentation* (47b):

For every possible world  $w \in \mathcal{O}'$ ,  $\{d \in D_w \mid person_w(d) \wedge \neg MP_w(d) \wedge PE_w(d)\} \neq \emptyset$   
 → Some people did not make a presentation but passed an exam.

The meaning contributions in (46) and (47) are the same, and the sets of contributive entities are part of the domains of the quantifiers.

## 6.2. Indicative Conditionals

Disjunction sentences and conditionals yield Gazdar's clausal implicatures. Here is an example of a conditional and its clausal implicatures.

- (48) If John is at home, he is safe.  
 +> John might be at home.  
 +> John might be safe.  
 +> John might not be at home.  
 +> John might not be safe.

In this paper, implicatures that arise from a disjunction structure are analyzed as EIs. Then it is expected that clausal implicatures from conditionals can be analyzed in the same way. This is really the case. To explain how, I first give the semantics of a conditional:

(Lewis's account)

'If  $\phi$ , then  $\psi$ ' is

- (i) vacuously true if and only if  $\phi$  is false;  
 (ii) non-vacuously true if and only if, among the worlds where  $\phi$  is true, some worlds where  $\psi$  is true are closer to the actual world than any world where  $\psi$  is not true;  
 (iii) false otherwise.

- (i) We didn't give every student of ours both a stipend and a tuition waiver. (Fox 2006)  
 +> There are SOME students of ours who we didn't give a stipend (but got a tuition waiver).  
 +> There are SOME students of ours who we didn't give a tuition waiver (but got a stipend).

Based on this truth-conditions, we can calculate the meaning contributions of the antecedent and consequent clauses of a conditional 'if  $\phi$ , then  $\psi$ '.

Meaning contribution of  $\phi$ :

$$\{w \in \sigma \mid \text{if } \phi, \text{ then } \psi \text{ in } w\} \uparrow \{w \in \sigma \mid \text{if Null}_{\tau(\phi)}, \text{ then } \psi \text{ in } w\}$$

$$\{w \in \sigma \mid \neg\phi \wedge \neg\psi\} \neq \emptyset$$

→ It is possible that  $\neg\phi \wedge \neg\psi$ . (MIGHT $\neg\phi$ , MIGHT $\neg\psi$ )

Meaning contribution of  $\psi$ :

$$\{w \in \sigma \mid \text{if } \phi, \text{ then } \psi \text{ in } w\} \uparrow \{w \in \sigma \mid \text{if } \phi, \text{ then Null}_{\tau(\psi)} \text{ in } w\}$$

$$\{w \in \sigma \mid \phi \wedge \psi\} \neq \emptyset$$

→ It is possible that  $\phi \wedge \psi$ . (MIGHT $\phi$ , MIGHT $\psi$ )

From the use of the antecedent clause  $\phi$ , we can get the EI that it is possible that  $\neg\phi$  and  $\neg\psi$ . That is, MIGHT $\neg\phi$  and MIGHT $\neg\psi$ . Similarly, we can get two EIs MIGHT $\phi$  and MIGHT $\psi$ , from the use of the consequent clause. Combined together, they are exactly the same as in (48).

## 7. Conclusion

In this paper, I compared EIs with SIs and showed they are different. When we get SIs, the relevant operator is negation, while EIs always have existential force. This indicates that the two implicatures arise due to different reasonings. EIs arise by the maxim of relation, while SIs by the maxim of quantity. Both EIs and SIs are sensitive to monotonicity, but monotonicity applies only in a limited context in generating EIs: between a disjunction structure and a quantifier which has immediately wide scope over it. EIs and SIs can arise locally, but they are basically global. Clausal implicatures are also EIs and this explains why EIs are more robust than SIs.

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